

**ION**

An atom that has lost or gained one or more electrons and as a result has either a positive or negative electric charge.

IONIC BOND

A type of chemical bond where one or more electrons are passed from one atom to another, creating two ions of opposite charge that attract each other.

ISOTOPE

One of two or more atoms of a chemical element that have different numbers of neutrons compared to other atoms of the element.

LIFT

The upward force produced by an aircraft's wings that keeps it airborne.

LIQUID

A state of matter that flows and takes the shape of a container, and cannot be compressed.

MAGMA

Hot, liquid rock that is found beneath Earth's surface.

MAGNET

An object that has a magnetic field and attracts or repels other magnetic objects.

MASS

A measure of the amount of matter in an object.

MATERIAL

A chemical substance out of which things can be made.

METAL

Any of many elements that are usually shiny solids and good conductors of electricity.

MICROORGANISM

A tiny organism which can only be seen with the aid of a microscope. Also known as a microbe.

MINERAL

A solid, nonliving material occurring naturally and made up of a particular kind of chemical compound.

MOLECULE

A particle formed by two or more atoms joined by covalent bonds.

MONOMER

A molecule that can be bonded to other similar molecules to form a polymer.

NERVE

A fiber that carries electrical messages (nerve impulses) from one part of the body to another.

NEUTRON

One of the tiny particles in an atom. It has no electric charge.

NUCLEUS

The control center inside the cells of most living organisms. It contains genetic material, in the form of DNA. Also: the central part of an atom, made of protons and neutrons.

NUTRIENT

A substance essential for life to exist and grow.

ORBIT

The path taken by an object, for example, a planet, that is circling around another.

ORGAN

A group of tissues that makes up a part of the body with a special function. Important organs include the heart, lungs, liver, and kidneys.

ORGANISM

A living thing.

PARTICLE

A tiny speck of matter.

PHOTOSYNTHESIS

The process by which green plants use the sun's energy to make carbohydrates from carbon dioxide and water.

PHYSICS

The science of matter, energy, forces, and motion.

PIGMENT

A chemical substance that colors an object.

POLLEN

Tiny grains produced by flowers, which contain the male cells needed to fertilize eggs.

POLYMER

A long, chainlike molecule made up of smaller molecules connected together.

PRESSURE

The amount of force that is applied to a surface per unit of area.

PRODUCT

A substance produced by a chemical reaction.

PROTEIN

A type of complex chemical found in all living things, used as enzymes and in muscles.

PROTON

One of the tiny particles inside an atom. It has a positive electric charge.

RADIATION

Waves of energy that travel through space. Radiation includes visible light, heat, X-rays, and radio waves. Nuclear radiation includes subatomic particles and fragments of atoms.

RADIOACTIVE

Describing a material that is unstable because the nuclei of its atoms split to release nuclear radiation.

REACTANT

A substance that is changed in a chemical reaction.

REACTIVE

A substance that is likely to become involved in a chemical reaction.

RESPIRATION

The process occurring in all living cells that releases energy from glucose to power life.

ROOM TEMPERATURE

A standard scientific term for comfortable conditions (for humans), usually a temperature of around 68°F (20°C).

SEX ORGANS

The organs of an organism that allow it to reproduce. They usually produce sex cells: sperm in males, and eggs in females.

SOLID

A state of matter in which an element's atoms are joined together in a rigid structure.

SOLUTE

A substance that becomes dissolved in another.

SOLVENT

A substance that can have other substances dissolved in it.

SYNTHETIC

Man-made chemical.

TISSUE

A group of similar cells that carry out the same function, such as muscle tissue, which can contract.

TOXIC

Causing harm, such as a poison.

ULTRASOUND

Sound with a frequency above that which the human ear can detect.

ULTRAVIOLET

A type of electromagnetic radiation with a wavelength shorter than visible light.

UNIVERSE

The whole of space and everything it contains.

VOLCANO

An opening in Earth's crust that provides an outlet for magma when it rises to the surface.

WAVE

Vibration that transfers energy from place to place, without transferring the matter that it is flowing through.

WAVELENGTH

The distance between wave crests, usually when referring to sound waves or electromagnetic waves.

WEIGHT

The force applied to a mass by gravity.